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Laboratory based Experimental Investigation of Photovoltaic (PV) Thermo-control with Water and its Proposed Real-time Implementation

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ABSTRACT

The output of Photovoltaic (PV) panel varies with panel temperature and in order to obtain uniform output from PV panels, water cooling system is recommended by several researchers worldwide. In this study, a laboratory scale experimental setup is developed (irradiance over the PV panel is varied from 87.38 W/m^2 to 359.17 W/m^2 and back to initial value in a triangular cyclic shape analogous to daytime radiation variation) for intermittent and continuous water cooling, which is tested for different flow rates. A mathematical analysis of temperature over PV panel with and without water cooling system is also done. For intermittent cooling case, three different flow rates of 3 lit/min, 5.3 lit/min, and 6.2 lit/min are used, which give an increase in total energy produced against no cooling operation of about 18% for all the three cases. The test results for continuous cooling (wherein a fixed flow rate of 0.6 lit/min was maintained), the total energy produced increases by approximately 29% against no-cooling operation. The proposed method of cooling the PV panels can be integrated with the water supply system of domestic dwellings. Further, a parametric study is carried out in order to choose the appropriate capacity of pump required for the operation. The proposed approach of integrating cooling of PV panels and water supply minimizes water loss (except evaporation loss) and reuses the cooling water back into the main water supply system of the dwelling. It is observed that the continuous cooling system with a water pump of appropriate capacity improves the performance of PV panels while keeping the consumption of water to minimum.

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1. Introduction

Due to fast depletion of fossil fuels, there is a need to improve the quality and efficiency of renewable energy technologies. Solar energy is the most accessible and widely deployed source in the form of various technologies such as: solar photovoltaic (SPV), solar air heater, parabolic trough collector, etc. Out of these technologies, solar PV technology is simple when it comes to installing and extensively used for domestic as well as large scale power generation. However, the performance of PV panel is dependent on the panel temperature [1–3]. It is reported in the literature that a temperature rise of 1 °C results in the loss of performance of PV panel by 0.5% [4–6]. It is also observed that the performance of solar PV panel is maximum when the temperature on the surface of panel lies in the range of 25 °C–35 °C [7–15] depending on the ambient conditions and the cooling technique adopted to curb this problem.

In the light of the above mentioned discussion, this paper focuses on improving the performance of PV panel using water as a cooling medium. A detailed review of literature is carried out to understand the influencing parameters, which affect the performance. Laboratory scale experimentation is performed to investigate the effect of cooling on the performance of PV panel and the results are presented along with a mathematical analysis. An optimized and easily adoptable scheme to implement such cooling system is also recommended.

1.1. Review of literature

Various techniques are used by researchers to reduce the

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internal thermal degradation of PV panel. These techniques include air ventilation system with natural [16–18] and forced type [19–21] circulation, heat pipe [22,23], thermoelectric cooling [24] and phase change materials [25,26]. The performance of PV panel can also be increased by allowing the water to flow over the surface of PV panel [27]. It not only reduces the surface temperature but also helps in decreasing the angle of incidence of radiation on the panel. In addition, it also acts as a cleansing agent and removes dust particles, which cause degradation in performance [7,11,28], from the panel. Krauter [12] reported a gain of 8–9% in efficiency when a thin layer of water was made to flow over the top surface of the panel during a clear day at the PV-Labs in Rio de Janeiro. The author also reported that the reflectance losses can be brought down to approximately 2–4% using a thin water film over the PV panel surface, which, in turn, reduces the surface temperature of the panel to 22 °C. Abdolzadeh et al. [7], investigated the possibility of improving the performance of a PV water pumping system at Mahan (Kerman) Renewable Energy (MRE) Lab by spraying water over the PV panels. An improvement of 12.5% in mean efficiency was recorded during the test day.

In order to simulate the performance of PV panel using a cooling medium, Kim et al., [9] developed a transient model using an energy balance of heat and mass transfer flow. Their model is capable of predicting cooling water temperature, PV module temperature, and evaporation rates. The model has predicted a power enhancement of approximately 12%. The authors have validated the model against experimental tests conducted on the roof of a three-storied building at Kangwon National University, Chuncheon, Korea. A maximum deviation of 4 °C was observed between the predicted and measured temperature of water and the panel. Working in the same direction, Moharram et al., [13] proposed a mathematical model using heating and cooling rate to determine the maximum allowable temperature on the panel at which the cooling of the panel should be started. Under the experimental conditions (German University in Cairo, Egypt) at which the tests were conducted, the maximum power output can be achieved only when the cooling starts at 45 °C. It was recommended that a water cooling system needs to be operated for every 5 min, which may result in 10 °C reduction in the solar cell temperature. Odeh et al. [14], carried out a long term performance of solar water pumping system under desert climatic conditions, HU University, Jordan. A gain of 15% in PV efficiency was observed during peak irradiation period with the application of water cooling. Using these test results with site radiation and ambient temperature data, a transient numerical simulation on a commercial software was carried out in order to evaluate the long term performance of the system. The simulation results indicated that using water cooling results in a steady increase in performance for dry and warm weather conditions, while there is a significant increase in performance for moderate weather conditions.

It is reported that the performance of PV panel is mainly dependent on its installation and associated environmental conditions [29]. In order to consider the effect of various conditions of cooling, Smith et al. [15], demonstrated the water cooling of PV panel at Portland State University (PSU) photovoltaic test facility where tests were conducted for open rack and insulated type PV modules; normal water and ice water cooling of PV panel; and combination of water cooling with concentrating methods on PV panel. In the case of open rack modules and insulated modules, operating on a particular test day condition, the net energy gain after accounting the pump power consumption was found to be 6% and 8%, respectively. An improvement of 24% in peak power produced by the PV panel was observed when ice water was allowed to flow over it for 1.5 h. In another set of experiments, use of V-trough reflector was made, while cooling is done simultaneously. In this experiment, it was observed that irrespective of the increased light intensity, the panel temperature was maintained at lower values with an improvement of 43% in peak power output. To study the effect of intensity of radiation over water cooling, Kolhe et al., [10] evaluated the performance of concentrated photovoltaic (CPV) system at the Southeast University, China. According to the results shown by the authors, using a CVP with water cooling can produce around 5 times more output electricity than that produced with a fixed PV panel.

As reported earlier, water cooling is a simple and efficient way of enhancing the performance of PV panels, but there are concerns like pump power requirement and reusability of the water flowing over the panel. Tripanagnostopoulos [30] proposed a dual heat extraction based hybrid photovoltaic/thermal (PVT) solar system that regulates the temperature of the PV panel to provide a higher thermal and electrical efficiency of the system. For the thermal control, either water or air can be used and the extracted heat can be utilized in fulfilling the building energy demands. Using air as a heat extraction fluid for active cooling, Teo et al., [31] studied the performance of a hybrid PVT system installed on the roof top at National University of Singapore and found out PV performance efficiency of 8%–9% and 12%–14% for the case of without and with cooling, respectively. Kordzadeh et al. [11], tested the performance of PV panels used for water pumping application over a period of four month at Kerman city. The part of water discharged from the pumping system was used for circulation over the surface of the PV panel. The effects of the nominal power of array and the system head on the operation of PV water pumping set were studied. It was found that the decrease in nominal power of array and subsequent increase in system head lead to increase in power generated by the array, which, in turn, increased the flow rate of water from the pump. As water flows over PV panel, heat is absorbed by cooling water through convection heat transfer, which increases the water temperature. This water with increased temperature can be used with the application of a heat exchanger. Hosseini et al., [8] studied an effective utilization of heat gained by the cooling water over 14 days trial in Tehran. It was found that the total output of the system was increased as compared to that obtained through a conventional system, due to waste heat recovery. This recovered heat can be further utilized for household applications. A maximum temperature of 18.7 °C and a relative difference in efficiency of 33% was observed, while the flow rate was maintained at 1 lit/min.

During the aforementioned review of literature, it is observed that several researchers have pointed out a need for better utilization, reuse and extraction of energy from the water that is used for cooling the PV panel. Therefore, unlike our preliminary results published in an international conference proceeding ICES-2015 [32], different flow rates of water are tested in the present study through several simulated laboratory experiments, which are made analogous to the real-time conditions. Also, an analytical heat transfer model is developed to predict the temperature over the PV panel with and without water cooling system. The tests were conducted using intermittent and continuous cooling approaches. The paper also reports the effects of cooling on the power produced from the PV panel at different flow rates using these approaches. Thereafter, a method of optimization, in relation to the installation of the fixed PV panel with water cooling system on the roof of residential buildings, has been proposed. Further, a parametric theoretical modelling approach is adopted to choose the water pump of appropriately size/capacity.

2. Mathematical analysis

Thermal analysis of the solar PV panel can be analysed in two different parts: without cooling and with cooling. A mathematical

model is developed to predict the temperature of the PV panel based on the irradiance and its heat transfer with the ambient or the cooling medium. Since the radiation is always normal to the PV panel surface for the present experimental conditions, there is no effect of orientation of the panel with respect to the ground. However, there will be a significant effect on the natural convection taking place over the PV panel surface, if the panel is oriented at an angle from the horizontal. For the experimental study, the panel surface is always kept at a horizontal position with the ground and thus, the same condition is assumed for the analytical model. Following are the components of heat transfer taking place over the PV panel:

a) Radiation falling on the panel surface, irradiance; $E = \alpha \cdot G_t$
b) Radiative heat loss between the panel and the ambient; $Q_r = \varepsilon \cdot \sigma \cdot (T_p^4 - T_{amb}^4)$
c) Natural convection heat loss from the panel; $Q_{N.\ cov.} = h_{NC} \cdot A_p \cdot (T_p - T_{amb})$
d) Forced convection heat loss (water based); $Q_{F.\ cov.} = h_{FC} \cdot A_p \cdot (T_p - T_w)$
e) Electrical DC power produced from the panel; $Q_{elec.} = C_{ff} \cdot \frac{E \cdot \log(K_1 \cdot E)}{T_p}$ [33]

Value of natural convection heat transfer coefficient (h_{NC}) is assumed to be 5.8 W $m^{-2}K^{-1}$ [34]. For the water flow rate varying from 0.6 lit/min to 6.2 lit/min used in the present study, the value of forced convection heat transfer coefficient is calculated with the help of standard correlation derived for laminar and turbulent flow over a flat isothermally heated plate [35]. The heat transfer coefficient is found to be in the range of 550 W $m^{-2}K^{-1}$ to 6500 W $m^{-2}K^{-1}$.

Without cooling:

$$\rho_p \cdot V_p \cdot C_p \cdot \frac{\partial T_p}{\partial t} = A_p \cdot (E - Q_r - Q_{N.\ cov.} - Q_{elec.}) \quad (1)$$

With Cooling

$$\rho_p \cdot V_p \cdot C_p \cdot \frac{\partial T_p}{\partial t} = A_p \cdot (\tau E - Q_r - Q_{F.\ cov.} - Q_{elec.}) \quad (2)$$

3. Experimental method

3.1. Experimental setup

In the present work, laboratory model based experiments are performed on a solar PV module (see Table 1) installed at BPHC, India as given in Fig. 1. The PV module includes two panels, of which, one is used for performing the experiment. A thermocouple is used at the back side of the panel to represent surface temperature. The panel is connected to a DC/AC convertor which is equipped with maximum power point tracking (MPPT) device. The complete set-up is connected to a data logger, which gives the current, voltage and power output values at the input of DC/AC convertor. The data thus obtained is stored in a computer system connected to the data logger. The experimental setup has a provision for varying the load on the system using a resistance pot. During the present analysis, a constant resistance of 115.5 Ω is applied. The required radiations are obtained from a set of 6 halogen bulbs (of rated capacity of 150 W each) fitted on the top of the panel. The variation in the intensity of radiation is observed using a pyranometer included in the setup.

To conduct different set of experiments, changes in the radiation values are obtained through the halogen light setup and variable voltage supply. Unlike the work of Irwan et al. [36], where an indoor water cooling based PV panel performance is tested at fixed average irradiation values, the derived irradiation variation scheme in the present study is analogous to the variation in daily solar insolation intensity. For this purpose, an analogy is developed representing

Table 1
PV panel specifications.

Maximum rated power	40 W
Operating Voltage	21.95 V
Short Circuit Current	2.44 A
Rated Voltage	17.84 V
Rated Current	2.25 A
Cell number	36
Area of Panel	0.24 m^2
Cell size	155 mm × 45 mm

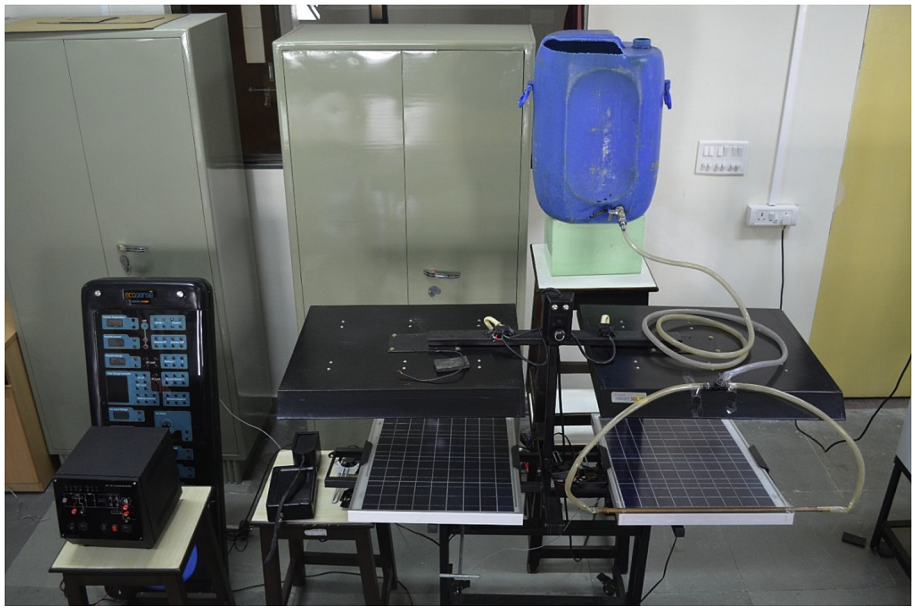


Fig. 1. Experimental setup.

variations in actual radiations with time. This is done by varying the supply voltage-current to the artificial light setup with respect to the recorded real-time data. According to this analogy, the average time taken for change of radiation intensity consecutively from lower to the peak and again from peak to the lower one is computed. Radiation intensity from the halogen light set-up is measured at nine different positions on the surface of panel and an average value with standard deviation (SD) is calculated corresponding to the set voltage. In this manner, irradiance value is being varied from 87.38 ± 17 W/m^2 to 359.17 ± 72 W/m^2 corresponding to the voltage variation from 85 V to 230 V. A reverse cycle of these values is taken to make the process analogical to the day time radiation variation. The whole experiment takes about 250 min to get completed as shown in Fig. 2. This radiation variation procedure is kept constant across all the experiments performed in the present study. A relatively high value of SD signifies the partial shading effect over the PV panel. It could have been contributed by the non-uniform illumination of the 6 halogens. As observed from the SD bars shown in Fig. 2, this effect increases with the increase in irradiance. Due to the partial shading effect, the PV module is expected to produce low power output than the rated power at a given irradiance for the laboratory setup used in the present study. In the real time, irradiance is usually found to be almost constant (with little or no variation) during the peak irradiance period. Hence, the middle portion of the plot in Fig. 2 is shown almost flat for the analogous lab experimental setup. Although, the standard testing procedure uses 1000 W/m^2 irradiance and 1.5 spectrum air mass at 25 °C operating temperature, but for the current laboratory based study, a peak irradiance value of 359.17 W/m^2 is maintained due to lux limitation of the halogen lights.

For the cooling purpose, a water tank of 50 liters capacity connected to the water supply line is used. It is an open circuit flow system, as the heated water from the panel surface is not being circulated back to the supply tank. The supply tank is placed in such a way that it provides a total head of 1 m to the water outlet at the top of panel surface. The water flow from the supply tank to the PV panel surface can be regulated with the help of a control valve. In order to maintain uniform distribution of cooling water on the surface of PV panel, a copper tube of length equal to width of PV panel with 8 holes of 1 mm each, is attached to the top of the panel as shown in Fig. 1.

3.2. Experimental procedure

Each experiment is started from an initial voltage of 85 V, which corresponds to the lowest value of the radiation observed on a typical day. Using the curve represented in Fig. 2, the change in irradiance is replicated. The corresponding surface temperature of the panel is recorded at fixed intervals of time. For the entire duration of the experiment average power is calculated by taking average at every 15 data intervals in the instantaneous power data derived from the data logger and plotted against time for all the experiments. To analyze continuous or intermittent cooling of PV panel, four different flow rates were maintained during the experimentation. For intermittent cooling, volume flow rates of 3 lit/min, 5.3 lit/min, and 6.2 lit/min are used, while a volume flow rate of 0.6 lit/min is used for continuous water cooling. In all the experiments for intermittent cooling cases, flow of cooling water automatically starts when the surface temperature reaches 40 °C and cuts off when the surface temperature decreases (or comes down) to 32 °C. It is observed that even though the flow of cooling water is stopped at 32 °C, the surface temperature of the panel drops down to the range of 27 °C–29 °C. However, it is observed that there is no substantial loss of power output due to this undercooling below 32 °C. Therefore, the water supply is cut-off at 32 °C. Moreover, for the case of continuous cooling, water is allowed to flow after the temperature on the panel reaches 40 °C and then stopped when irradiance has reached a corresponding value of 115 V during the reverse cycle of voltage variation.

4. Results and discussion

Three different sets of experiments are performed using the approach described in the previous section. In the first set of experiments, water cooling is not provided to the PV panel, while performing the other two sets of experiments, differential rates of water flow over the PV panel were maintained. As discussed earlier, the surface temperature of PV panel plays an important role in power output; therefore during these experiments, the water flow rate over the PV panel surface is varied and is considered to be an important variable during the analysis. This type of analysis helps in comparing the results for two different cooling methods performed at different flow rates, which are discussed in the following sections.

4.1. Without cooling

In order to draw inferences, three experiments were performed at standard operating conditions in the laboratory on three days so as to check the consistency in output variables. The results thus

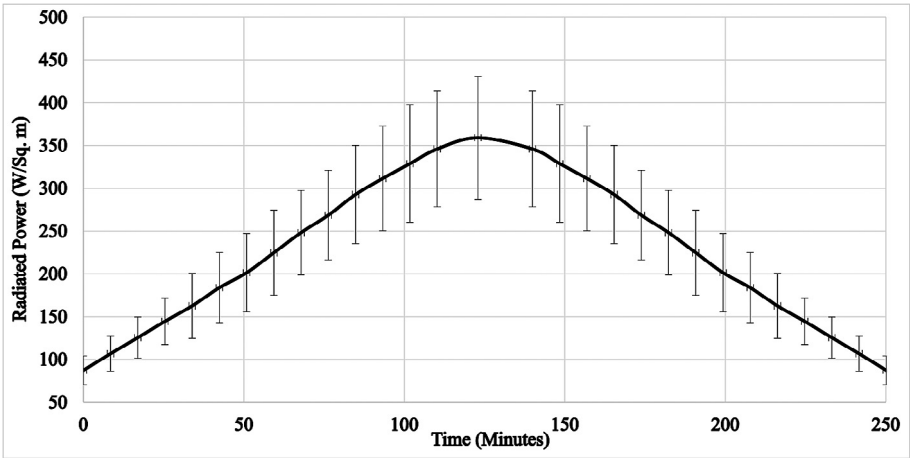


Fig. 2. Plot for radiated power with standard deviation bars from the halogens versus time for experiment.

obtained have very good consistency with respect to PV panel surface temperature and associated power output. It can be noted from Fig. 3 that the temperature over the PV panel surface reaches a maximum of 55 °C; with most of the time during the test run, it remains above 35 °C. As per the analytical model represented by equation (1) for the heat transfer over the panel, the obtained curve is plotted with the experimental values. Excluding the start and intersection points in Fig. 3, the absolute minimum and maximum difference between measured and model predicted temperature is found to be 0.5 °C and 4 °C, respectively. In Fig. 4, the model predicted temperature values are plotted against experimentally measured temperature values for the whole experiment duration and a linear regression model is applied on the scattered data so obtained. The linear regression correlation coefficient value (R^2) is found to be 0.9363. Further, standard error of estimate between model predicted and measured temperature is calculated to be 2.52 °C for the overall duration of the experiment. It should be noted that the experimental temperature data is obtained by varying the irradiance in multiple steps of varied time-step size following the trend shown in Fig. 2. As a result, the initial values of the two curves in Fig. 3 attain different slopes and lead to under prediction of temperature by the model. Moreover, at the instant of 112 s, there is a sudden rise in the slope of experimental temperature curve, which corresponds to the peak radiation zone. This results in a sudden increase in radiative heat transfer over the PV panel surface that is reflected in Fig. 3. Another reason for the deviation in two curves is the varied ambient conditions during the experiment, which is assumed to be constant for the analytical model.

Fig. 3 shows that the PV panel can produce a maximum average power output of 1.94 W at 43 °C surface temperature. It can also be noted that there is a drop in average power output with increase in surface temperature of PV panel. For most of the time during experimentation, the PV panel has produced average power output of less than 1.9 W due to increase in PV panel surface temperature. By integrating the power curve with respect to time in hours, the total energy produced throughout the experiments without any external cooling (except natural convection) is found to be 7.14 Wh.

As shown in Fig. 5, it is observed that the average power output reaches the maximum value when the PV panel surface temperature rises to 40 °C and remains at its peak till the temperature reaches 44 °C. Thereafter, a gradual decrease in average power is observed. In lieu of any cooling applied, a similar trend is observed for two other experiments and hence, in order to achieve a maximum power output, water cooling is applied over the PV panel surface. Under any cooling method adopted, the flow of water over the PV panel surface is started at 40 °C, so as to avoid additional thermal gradients. If this cooling is started beyond 40 °C, loss of water due to evaporation would also increase. Therefore, for the present work, the point where temperature over the PV panel surface reaches 40 °C is defined as the *"power saturation point"*.

4.2. With intermittent cooling

In order to improve the performance of PV panel, intermittent cooling system is utilized in experiment 2. Three different experiments were performed in this case with three different flow rates to vary the cooling rate of PV panel surface. Also, different flow rates account for the amount of the water required to achieve the required cooling rate. It is obvious that the higher the flow rate, the less is the time required to bring the surface temperature of the PV panel down. For the present study, the highest possible flow rate used as per the setup limitation is 6.2 lit/min. Further, in order to investigate the correlation between the flow rate on the cooling and power generation by PV panel, two more lower flow rates of 5.3 lit/min and 3 lit/min are also studied. Fig. 6 shows the variation in power produced and surface temperature of PV panel with respect to the experimental time duration. A cyclic heating and cooling of the PV panel is observed, starting from the moment when the PV panel surface temperature reaches 40 °C for the first time i.e. *power saturation point*. This 40 °C temperature is then brought down to 32 °C in the cooling process and hence follows the same thereafter. This heating and subsequent cooling continues during the entire duration of experimentation. Also, it can be observed that this heating-cooling cycle is more frequent in the middle of the experimentation duration. This is mainly because of the corresponding higher radiation observed on the PV panel during the middle phase of the experimentation duration.

As shown in Fig. 6, the minimum average power output from the PV panel is 2.05 W, which is almost equal to the maximum average power produced in without cooling case. The average total energy produced for the whole duration of experimentation is 8.42 ± 0.042 Wh for all the three experiments performed at three different cooling water flow rates. Using the standard statistical two-tailed student's t-test [37] on any two out of three power output data, the power produce for the three experiments is found to be significantly similar with a p-value of almost zero. It can be concluded that irrespective of the different cooling rates for different water flow rates, the power curve for all the three cases

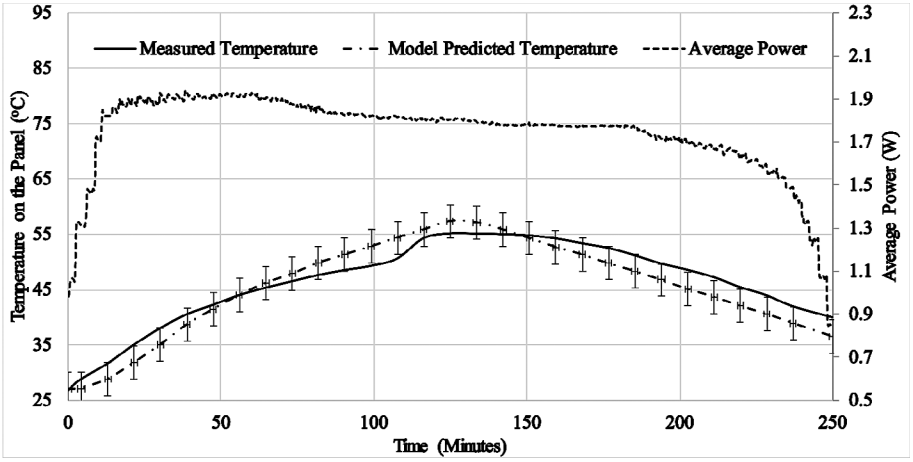


Fig. 3. Temperature on the PV panel with error bars within 3 °C maximum deviation and power produced for without cooling experiment.

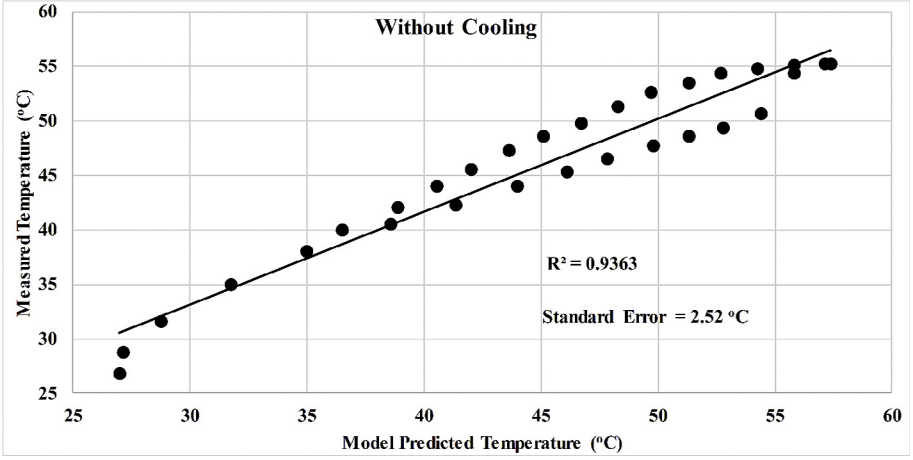


Fig. 4. Linear regression curve with correlation coefficient (R^2) and standard error for the measured and model predicted temperature over the PV panel (Without Cooling Experiment).

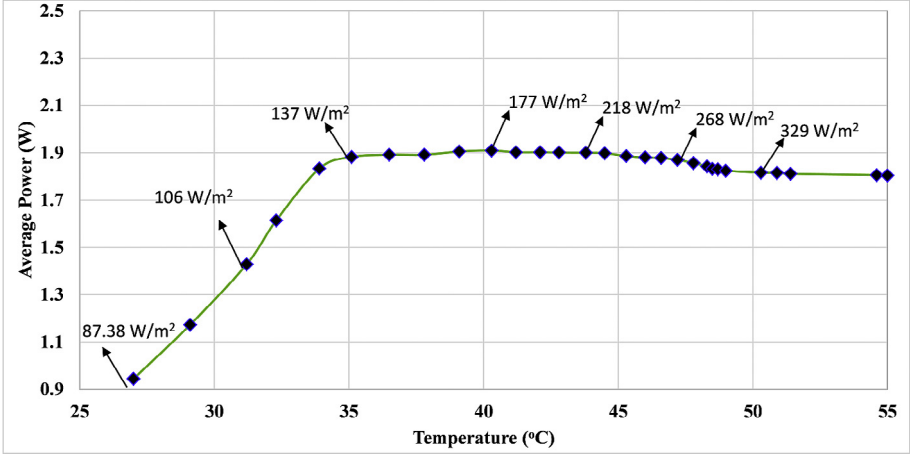


Fig. 5. Temperature versus power produced for without cooling experiment.

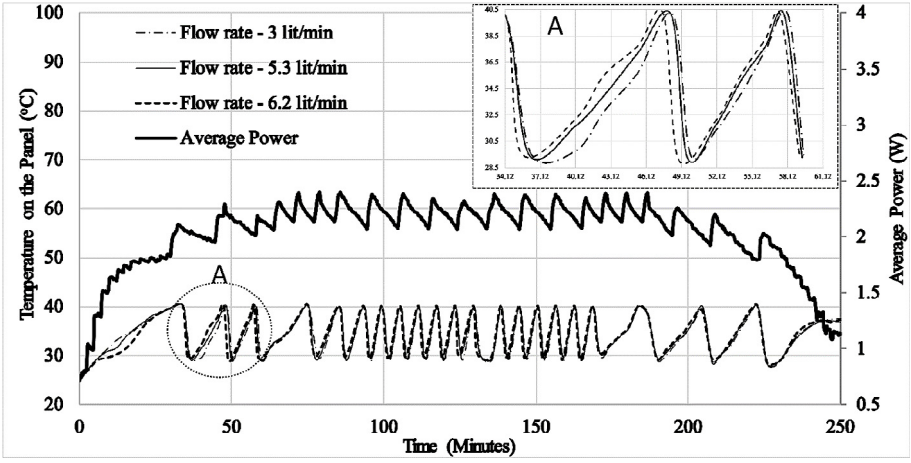


Fig. 6. Temperature on the PV panel surface for different flow rates and power produced during intermittent cooling experiment.

follows a similar trend, which encourages the authors to include only one of those curves as shown in Fig. 6. This has further restricted the authors to test any further lower water flow rates for intermittent cooling. It is observed that the maximum average power that could be achieved through this method of cooling is 2.45 W. While comparing the performance of PV panel under intermittent cooling versus no cooling condition, an increase of approximately 17.9% in total energy produced is observed. The total

amount of water required and the total time for cooling for all the three cases are summarized in Table 2. Model predicted cooling rates during peak radiation are also summarized in Table 2, which are slightly deviated from the observed experimental cooling rates. Possible reasons for this deviation could be the variability in the cooling water supply temperature and ambient conditions during the experiment; which are considered constant for the analytical modelling.

In order to implement an intermittent cooling system in a real life situation, an electro-mechanical device is required to regulate the flow rate of cooling water with time. This system includes a temperature sensor that feeds its output into a microcontroller. Based on the temperature inputs, the microcontroller controls a servo-motor to operate the flow control valve as per the flow rate requirements. This process is repeated and operated as and when the temperature on the PV panel surface reaches its *power saturation point* (from which cooling should be started). However, this type of cooling system is energy (electrical) intensive and requires some routine maintenance. Moreover, such a solution is the worst in terms of water requirement in the sense that it encourages water wastage and increases water pump power requirement.

On the contrary, if the system is designed with a constant and continuous flow rate then possibly it will mitigate all these associated problems. Also, if the water flow rate is kept to a minimum possible value while ensuring maximum spread over the PV panel surface, then this can be a feasible solution. Therefore, in the next set of experiments, a continuous cooling method with a minimum possible cooling water flow rate (for the designed experimental setup) is tested, which is discussed in detail in the next section.

4.3. With continuous cooling

In this case, the experiment is done for continuous cooling water flow rate of 0.6 lit/min throughout the cooling duration. As shown in Fig. 7, cooling is started after the surface temperature of PV panel has reached its *power saturation point*. After starting the cooling, first the temperature is dropped to a minimum value of 25 °C, but as the radiation intensity and cooling water supply temperature (not controlled) vary, they affect the effective heat transfer over the PV panel and so the PV panel maintains its temperature between 31 °C and 32 °C. However, for the analytical model presented, cooling water supply temperature is considered to be constant (=25 °C) throughout the duration of experiment which is the ideal cooling scenario. Therefore, the model predicted curve (using equation (2)) shows a slight temperature variation corresponding to the irradiance change. For this case, the absolute minimum and maximum difference between measured and predicted temperature is found to be 0.8 °C and 4.4 °C, respectively. Standard error of estimate between the measured and predicted temperature for the whole cooling duration of the experiment is calculated to be 3.2 °C. A similar deviation in the temperature is also reported in the work of Kim et al., [9]. If the time rate of change of cooling water supply temperature is known, then more accurate model prediction values can be evaluated.

Table 2
Summary of the experimental parameters for intermittent cooling.

Flow rate (lit/min)	Total time for cooling (min.)	Total amount of water (lit)	Cooling rate (°C/min.) expt. based (at peak radiation)	Cooling rate (°C/min.) model predicted (at peak radiation)
3.0	42.72	128.16	3.52	3.70
5.3	38.48	203.94	4.28	4.93
6.2	34.28	212.54	6.40	7.40

The power curve for this case is quite significant in terms of production of constant power by the panel. As shown in Fig. 7, the maximum average power that can be reached through this method is 2.6 W, and then it decreases by 8.33% and then, remains constant for a while. The decrease in the average power produced in this case is not as steep as that seen in the case of intermittent cooling. Moreover, at the end of the experiment a value of 1.7 W is achieved, which is higher than what is obtained in the intermittent cooling case. The total energy produced during the experiment is found to be 9.24 Wh, which accounts to approximately 29.40% increase in total energy produced as compared to without cooling case and approximately 9.70% increase over that seen in the case of intermittent cooling method. Moreover, the whole cooling duration for this case is 190 min, which is much higher than that in any of the three cases of intermittent cooling considered in the present study. However, the total amount of water spent on continuous cooling method is around 114 liters, which is about 11% less than the minimum and about 46% less than the maximum amount of water required in intermittent cooling with the lowest and the highest flow rates adopted in the present study, respectively.

5. Analysis of test results

5.1. PV panel performance versus temperature analysis

In this section, a comparative analysis of the effect of cooling and its absence on the PV panel is presented, respectively. Fig. 8 shows the average power output Vis-a'-Vis temperature of the PV panel surface. Change in average power output with two different methods of cooling when the temperature reaches to 40 °C (*power saturation point*) for the first time and thereafter, is compared. For analysis purpose, this point is superimposed to each other irrespective of slightly different irradiance values as an experimental limitation. It can be observed that in the absence of cooling, the power output increases from 27 °C to 40 °C and then remains constant for further increment of 3–4 °C. After this point, again, there is an increase in the temperature of PV panel surface with increase in irradiation (marked with arrows in Fig. 8), which leads to a substantial decrease in average power produced. Therefore, while adopting intermittent cooling, cooling water is allowed to flow over the surface of PV panel as soon as the temperature over the PV panel surface reaches its *power saturation point*. This flow of water brings the temperature of the PV panel down to 29 °C, which leads to an upsurge in the average power output of the PV panel. As the nature of water flow is intermittent, with the start of water flow at 40 °C, the temperature drops to 29 °C at irradiation value of 182 W/m^2 with an increased average power output as compared to without cooling case for the same irradiation. Thereafter, as the water flow stops, the PV panel starts gaining the temperature. At this point, the average power versus temperature curve starts returning through same line with a gradual deviation (Fig. 8) as the irradiation increases causing more power output at 205 W/m^2 and 210 W/m^2. As the temperature increases, the average power produced starts decreasing, however, it is still higher than without cooling case for the same range of irradiation as the temperature remains below the *power saturation point* before the next cycle of cooling is applied. This results in subsequent heating and cooling of the PV panel surface with a higher average power output compared to without cooling case under same irradiation. In case of continuous cooling method, flow of cooling water starts at the *power saturation point* and then allowed to flow over the surface of PV panel for the whole duration of the experimentation. As shown in Fig. 8, as the water starts flowing over the PV panel under irradiation of 177 W/m^2 and thereafter, the average power output

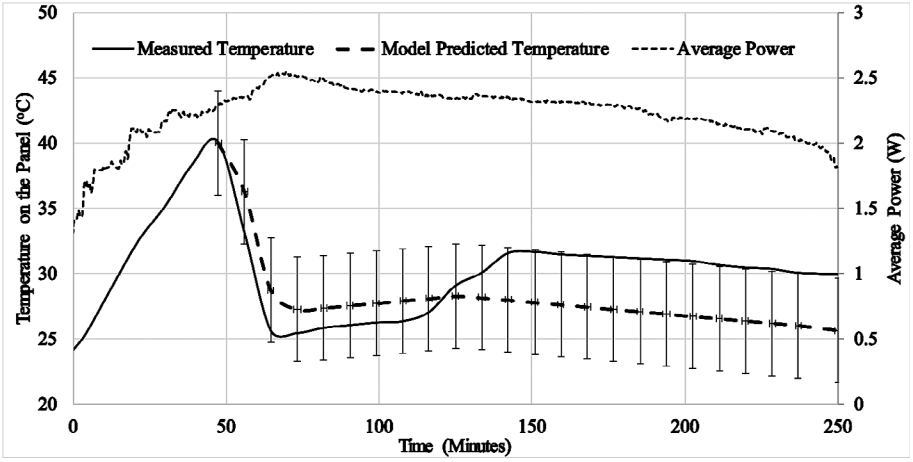


Fig. 7. Temperature on the PV panel with error bars of 4 °C maximum deviation and power produced for continuous cooling experiment.

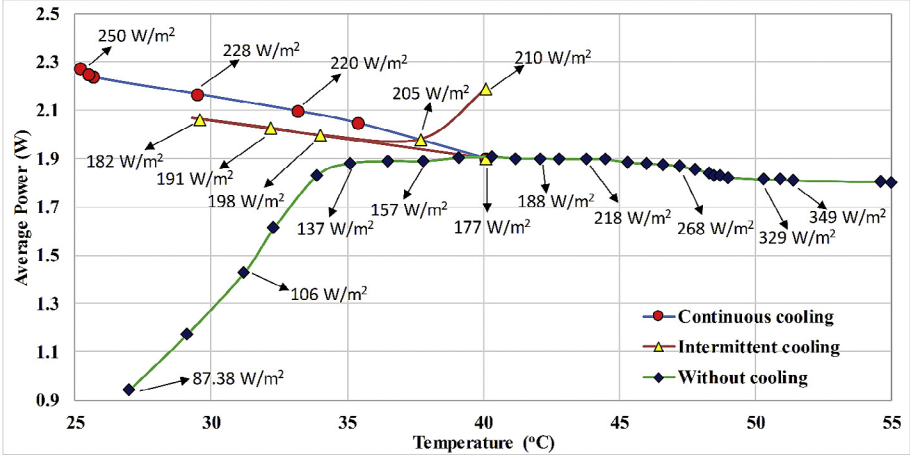


Fig. 8. Power versus temperature of the PV panel for different sets of experiment.

increases with the decrease in temperature and reaches a maximum value at 25 °C (250 W/m^2). A corresponding irradiation range leads to a decrease in average power output as the temperature increases (Fig. 8) for without cooling case. With this method, a continuous boundary layer can be achieved over the surface of PV panel, which helps in maintaining lower temperature over the PV panel and decreases the evaporation water losses as compared to that achieved by the intermittent cooling method. Moreover, referring to Figs. 3, 6 and 7, average power output during the peak radiation time and same temperature (around 29 °C) is observed to be higher in case of continuous cooling method (31.28% against no cooling) than that noticed in case of intermittent cooling (17.32% against no cooling).

5.2. Without cooling versus continuous cooling based PV panel performance analysis

In this section, a comparative analysis is shown between the overall performances of the PV panel under continuous cooling against no cooling. For this purpose, five different points on the power curve (same time instant and irradiance) are chosen at which average power output and its percentage increment are summarized in Table 3. It can be observed that most of the time during the experimentation, a minimum gain of 23% in power is achieved through continuous cooling of PV panel.

6. Proposed installation methodology

The worldwide capacity of solar PV panels has been raised from 5 GW in 2005 to around 140 GW in 2013 [38] and expected to increase further in subsequent years. Owing to the huge demand and temperature limitation encountered, an optimized and low cost solution for higher performance of the PV system needs to be investigated. As discussed in the previous sections of this paper, an appropriate water cooling system can improve the performance of PV panels. In order to provide cooling solution to PV panels, residential areas make a good choice, with availability of water at the roof top for daily household use. The major concern with the use of water for cooling of PV panel is the power consumed in the pumping system, which needs to be in accordance to the performance enhancement of the system. Therefore, there is a need to design a system, which maximizes the net benefit from the PV panels integrated with cooling techniques using minimum pumping energy input. Wu and Xiang [39] propped a strategy to use a gas expansion based system to spray rain water over the PV panel. The gas expands in proportion to the solar irradiation collected by the device that accordingly sprays the water over the PV panel. Employing the numerical simulation tool, the authors have showed an increase in the electrical efficiency of the PV panel by 8.3% with a payback period of less than 14 years. Similarly, a novel system design with efficient cooling capacity is proposed through the

Table 3
Comparison between without and with continuous cooling PV performance.

	Without cooling			Continuous cooling		
Points	Time (min.)	Temperature (°C)	Instantaneous Power (W)	Temperature (°C)	Instantaneous Power (W)	% Gain in power
1	45	40	1.96	37	2.25	14.52
2	70	46	1.97	25.7	2.45	24.86
3	125	55	1.88	30	2.36	25.33
4	200	47	1.80	31.2	2.20	23.49
5	250	40	1.20	30	1.75	45.89

present research work. Fig. 9 shows the schematic layout of the proposed scheme.

A residential building of height 'h' m with a water tank 'A' of capacity 1000 liters and height 't_A' m, installed on the roof top is considered. The tank is assumed to be placed at a height of 'a' m from the roof surface. Water tank 'B', of capacity 500 liters and height 't_B' m, needs to be placed on the surface of the roof. The solar PV panel is required to be installed at a height more than 't_B' m from the roof surface, with proper orientation based on the latitude and longitude of the location to maximize the solar gain. Typically, there could be two situations wherein either the main water supply tank could be strictly on the roof top or at a certain height from the roof top surface. From the structural balancing point of view, the overhead supply tanks are normally constructed at a height of 1–2 m above the roof top surface. Now, for cooling the PV panel, water will be taken from the tank 'A' and made to flow over the panel surface. While flowing over the panel, water will absorb the heat from the PV panel surface and cool it down. The water, which leaves the PV panel surface, will be collected in tank 'B'. Water in tank 'B' can be used for the household applications, where the temperature of water is not a matter of great concern, like wash-rooms, laundry, etc. For this purpose, a separate pipeline, other than the usual one coming from tank 'A' to different floors of the residential building, may be required.

As per the proposed scheme, there is no extra water pump, other than the main water pump to fill tank 'A', will be required for the cooling system. If the tank 'A' is intentionally installed at such a height that would provide a net head difference of 1.5 m (($a+ t_A$) - t_B), then the extra pumping energy required to pump the water at 'H' in order to run the whole system can be calculated from a parametric study as discussed hereafter. The main variable for this analysis is the height of the residential building, which usually varies from 10 to 20 m depending on the number of floors. From this parametric study, the excess pumping energy required to pump 500 liters of water at two different heights (with a net head difference of 1.5 m – one at 'H' and the other at '($h + t_B$)') can be expressed as a function of height 'H'. For this purpose, three different design models (A: PEARL; B: STAR-I, C: SUPERSTAR) of similar rated (0.37 KW capacity) water pump from Kirloskar pump catalogue [40] are considered. Data on pump discharge rate (lit/hr) as a function of pumping head is extracted from the catalogue. Energy required (in KJ) to fill 500 liters of water is calculated for a range of pumping head from 7 m to 20 m. Further, excess energy required for total head of 'H' corresponding to a net head difference of 1.5 m is calculated and plotted for the three different pumps as shown in Fig. 10(a).

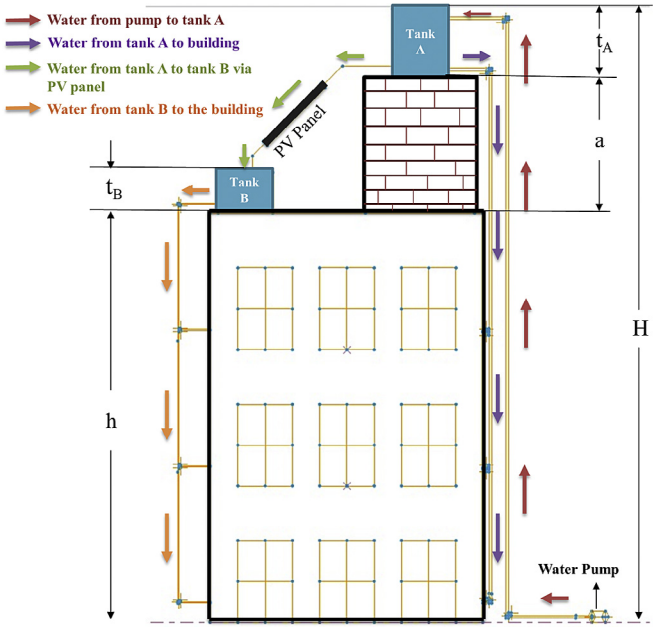


Fig. 9. 2-D Sketch for Installation of water system for cooling of the PV panel on a residential building.

For the same rated power pump, as the pumping head increases, discharge rate from the pump decreases. Conversely, it would take more time (results into more electrical energy consumed) for the tank to fill at higher pumping head. However, if for a given pump, the excess energy required to pump at an additional height of 1.5 m doesn't vary much, it would suit the proposed application. It can be easily concluded from Fig. 10(a) that pump C is the best choice for this application, where the difference in excess energy required with change in head is not varying much. If the heights are H = 15 m, ($h + t_B$) = 13.5 m and diameter of the delivery pipe is 19 mm, then the flow rate of water from two tanks 'A' and 'B' at the bottom of the building (by using equations (3) and (4)) will be 4.86 lit/sec and 4.61 lit/sec, respectively.

$$Q = A \cdot V \tag{3}$$

$$V = (2 \cdot g \cdot h)^{\frac{1}{2}} \tag{4}$$

Therefore, the difference in the water flow rate coming from either of the tank and at a certain height of the building is found to be almost negligible (i.e. 0.25 lit/sec). A similar observation can be made from Fig. 10(b), where net head difference is 3 m. In this case, for H = 15 m, the difference in flow rate is 0.51 lit/sec, which can be appropriately neglected for household application with other head losses. It must be noted that for these calculations, only vertical head is considered and head loss due to friction, bends, cross-section, etc. are not taken into account.

7. Conclusions

In the present work, three different laboratory based experiments were performed on solar PV panel with and without cooling medium (water). The value of irradiance is varied from 87.38 W/m^2 to 359.17 W/m^2 and again back to the initial value in a triangular cyclic shape to perform a 250 min experiment. It is observed that

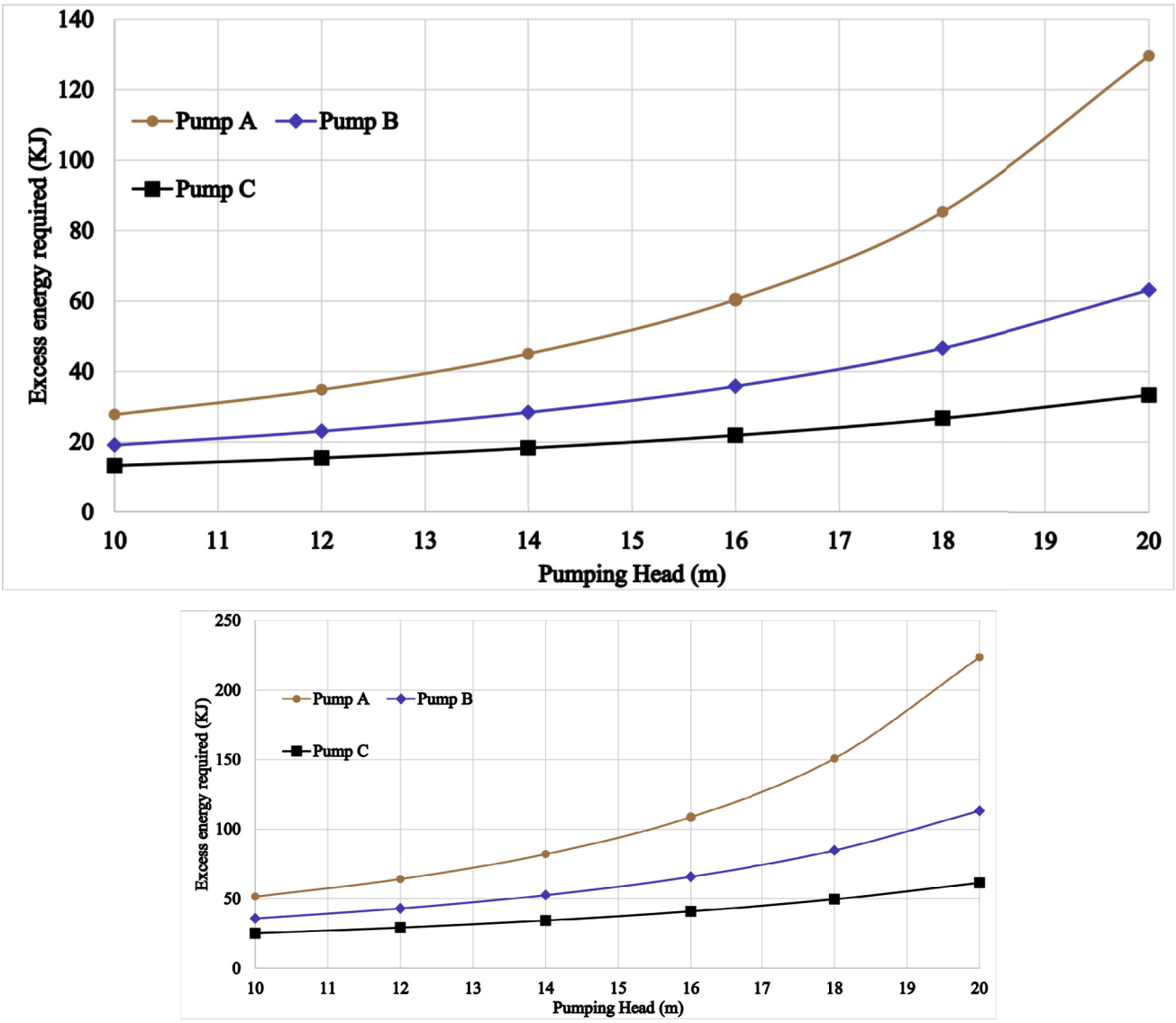


Fig. 10. (a): Extra pumping power versus pumping head 'H' with net head difference = 1.5 m. (b): Extra pumping power versus pumping head 'H' with net head difference = 3 m.

the average power output of PV panel is higher in case of continuous cooling than that in case of intermittent or no cooling. The difference in total energy produced between the two cases – continuous cooling and no cooling- amounts to almost 29.40%. The main reason for this difference is better heat absorption rate by the cooling water flowing over the PV panel surface in a continuous manner. Therefore, based on the lab experiments, a continuous cooling method is recommended and preferred over the intermittent and no cooling operation. In support of the method adopted, an optimized system is proposed for its application in the residential buildings. The main conclusions drawn out of this work are:

a) The maximum temperature that can be reached over the PV panel is 55 °C, while cooling is done from 40 °C up to 32 °C, which is supposed to be the optimal working temperature of the PV panel under tested conditions.
b) Two different methods for cooling are tested: intermittent cooling and continuous cooling. Against no cooling for which total energy produced is 7.14 Wh; for the former method, the gain in total power produced is 1.28 Wh, while for the latter method; the net gain is 2.1 Wh.
c) The quantity of water required for the continuous cooling method is less (i.e. 114 liters) than that for the intermittent cooling method, where minimum requirement is 128 liters, for the tested experimental duration.
d) Through the proposed installation scheme of the cooling water supply system with the PV panel, the input pump energy is very low, which is supposed to be easily compensated against the improvement in the performance of the system.

The results obtained at the laboratory scale investigation are encouraging to carry out similar studies in the outdoor condition.

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Nomenclature

α	Radiation absorptance of the PV panel (=0.75)
τ	Radiation Transmittance of water (=0.80)
G_t	Irradiance from the halogen lamp (W m^{-2})
ε	Emissivity of the PV panel (=0.95)
σ	Stefan Boltzmann constant (=5.6697 × 10^{-8} W m^{-2} K^{-4})
T_p	Temperature of the PV panel (K)
T_{amb}	Temperature of the ambient air (=298 K)
T_p	Temperature of the PV panel (K)
T_w	Temperature of the Water (=298 K)
h_{NC}	Natural convection heat transfer coefficient between the PV panel and the ambient air ($Wm^{-2}K^{-1}$)
h_{FC}	Forced convection heat transfer coefficient between the PV panel and the water ($W.m^{-2}.K^{-1}$)
A_p	Top surface area of the PV panel (=0.24 m^2)
C_{ff}	Fill factor model constant (=1.22 K m^2)
K_1	Constant = K/I_o (=10^6 m^2/W)
ρ_p	Density of PV panel material
V_p	Volume of the PV Panel
C_p	Specific heat capacity of the material of PV panel
Q	volume flow rate of water (m^3/s)
V	Velocity of flow (m/s)
g	acceleration due to gravity (=9.81 m/s^2)
h	height or head (m)

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